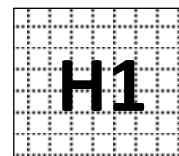


Civics Group	Index Number	Name (use BLOCK LETTERS)
--------------	--------------	--------------------------



ST. ANDREW'S JUNIOR COLLEGE
2013 Preliminary Examination

H1 BIOLOGY

8875/2

Paper 2: Core

Thursday

29 August 2013

2 hours

Additional Materials: Answer Paper
Cover Sheet for Section B

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** the questions.

Section B

Compulsory question to be answered on writing paper provided.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/11
2	/8
3	/8
4	/13
5/6	/20
Total	/60

This document consists of **13** printed pages.

[Turn over

Section A

Answer **all** questions.

QUESTION 1

(a) Fig. 1.1 is an electron micrograph showing part of a pancreatic cell.

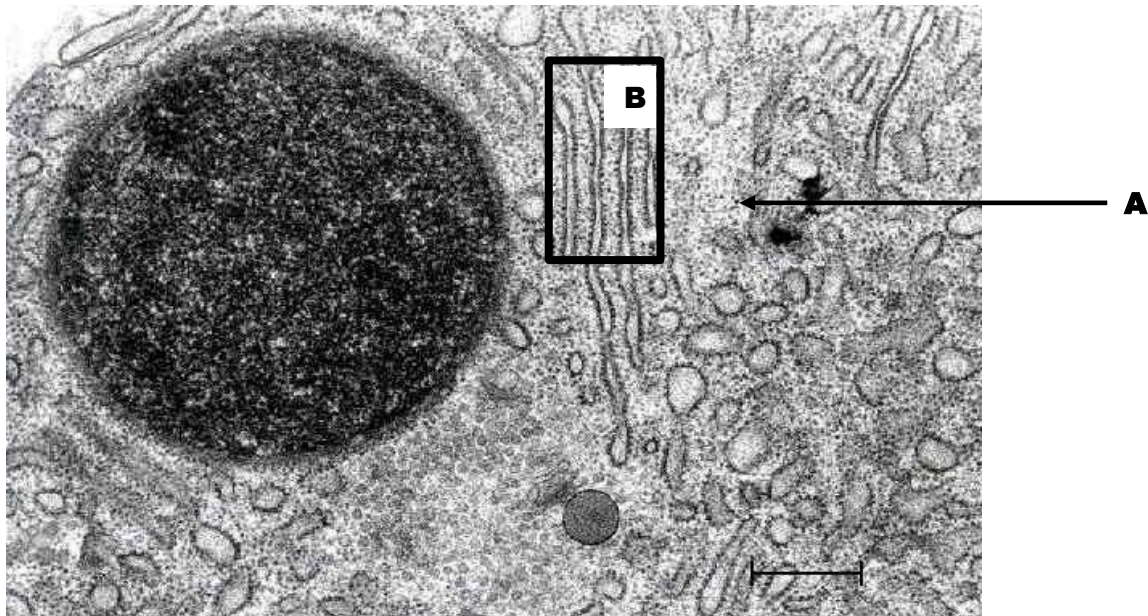


Fig. 1.1

(a)(i) Identify organelle A and describe its specific roles in the production of protein.

.....

.....

.....

.....[2]

(ii) Certain mitochondria inhibitors can inhibit the functioning of stalked particles. Explain the effects of such inhibitors on organelle B.

.....

.....[1]

(b) (i) Certain proteins were specifically imported into organelles such as chloroplasts. Suggest and explain how the membrane structure of chloroplast facilitates this process.

.....

[2]

(ii) An aqueous solution contains small membranous vesicles which may be derived from a thylakoid membrane. In these vesicles, the orientation of the ATP synthase enzymes is preserved as it is in the cell.

Fig. 1.2 shows the various experimental systems that are prepared in an attempt to synthesize ATP.

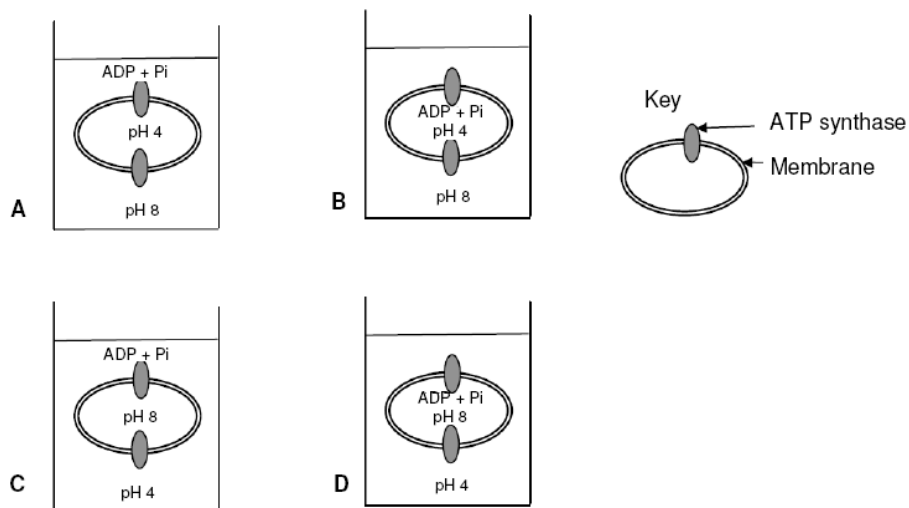


Fig. 1.2

Which experimental system (A to D) shown in Fig. 1.2 would result in the successful synthesis of ATP by thylakoid-derived vesicles? Explain your answer.

.....

[3]

(c) Figure 1.3 shows the effects of increasing substrate concentration on the rate of enzyme reaction for an enzyme functioning in the chloroplast.

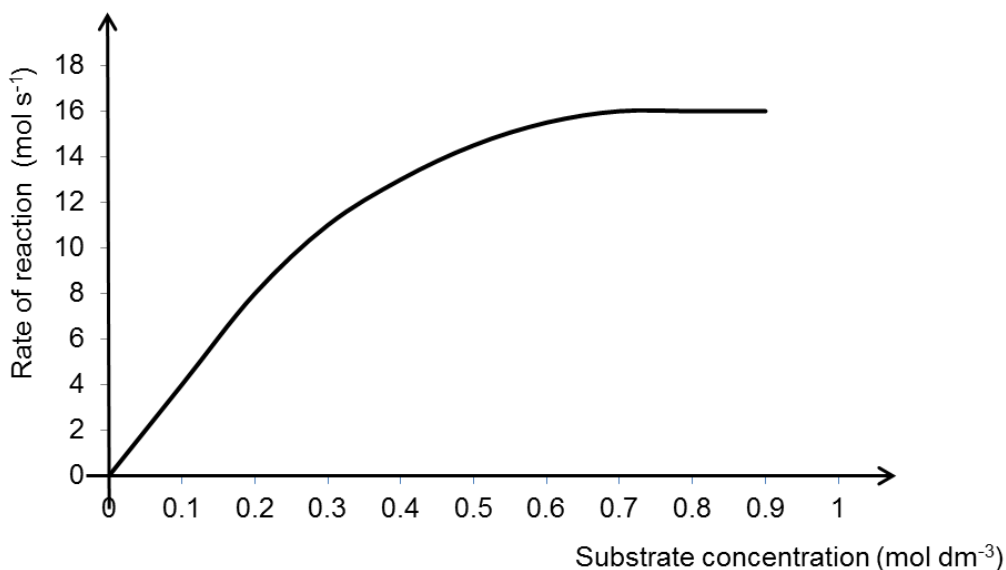


Fig. 1.3

(i) With reference to Fig 1.3, account for the rate of reaction from 0 to 0.3 mol dm⁻³ of substrate concentration.

.....

[2]

(ii) Exposure of the enzyme to chemical compound X affects the rate of reaction negatively. Analyses revealed that compound X has a shape complementary to a domain away from the active site, and has a low level of structural similarity between actual substrate and compound X. **Draw** on Fig. 1.3, the prediction of the rate of reaction of the enzyme after subjection to chemical compound X.

.....[1]

[11 marks]

QUESTION 2

Bt crops are named after *Bacillus thuringiensis* (*Bt*), a bacteria that naturally produces a crystal protein that is toxic to many pest insects. *Bt* crops are genetically engineered to produce the *Bt* toxin in every cell of the plant, with the goal of protecting the crops from pests.

To kill pests, ingested *Bt* toxins must first become activated by gut protease, before binding to a toxin-binding cadherin protein in the midgut. This cadherin protein is normally localized in the membrane and functions as a receptor for the toxin.

Mutation of the protease gene and/or the cadherin gene in the pests can result in a phenotype resistant towards *Bt* crops. Both genes are inherited in a recessive manner.

(a) Explain how these mutations result in resistance to *Bt* crops.

.....

.....

.....

.....[2]

(b) Define the following terms:

(i) genotype

.....

.....[1]

(ii) phenotype

.....

.....[1]

A cross between a *Bt* resistant insect (with mutant receptor, inactive protease) and a *Bt* susceptible insect will have a 75% chance of getting a *Bt* resistant offspring. Assume the insects are diploid.

(c) Using the symbols provided below, draw a genetic diagram to show these results.

.....[3]

Symbols:

Allele A: dominant allele for active protease

Allele a: recessive allele for inactive protease

Allele B: dominant allele for normal cadherin protein

Allele b: recessive allele for abnormal cadherin protein

(d) State one social/ethical concern, other than pest resistance, pertaining to the use of Bt crops in agriculture.

.....
.....[1]

[8 marks]

QUESTION 3

Fig. 3.1 shows a diploid germ cell with 3 pairs of homologous chromosomes during metaphase.

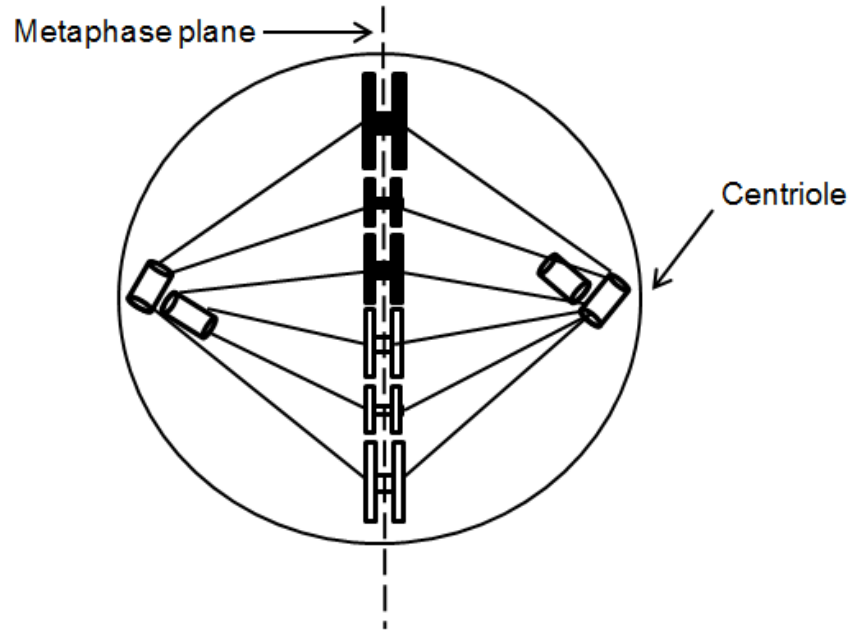


Fig. 3.1

(a) Explain why the homologous chromosomes in Fig. 3.1 do not form bivalents.

.....
[1]

(b) With reference to an appropriate property of stem cells, explain if stem cells undergo meiosis.

.....

[2]

(c) An error occurred **once** during gamete formation in a diploid germ cell of a woman. As a result, her children will have 50% chance of having trisomy of chromosome 9; and 50% are monosomy (only one copy is present) for chromosome 9 in his/her cells. Explain how this could have happened.

.....

[2]

Fig 3.2 shows a midsection of a DNA molecule during the S phase of interphase.

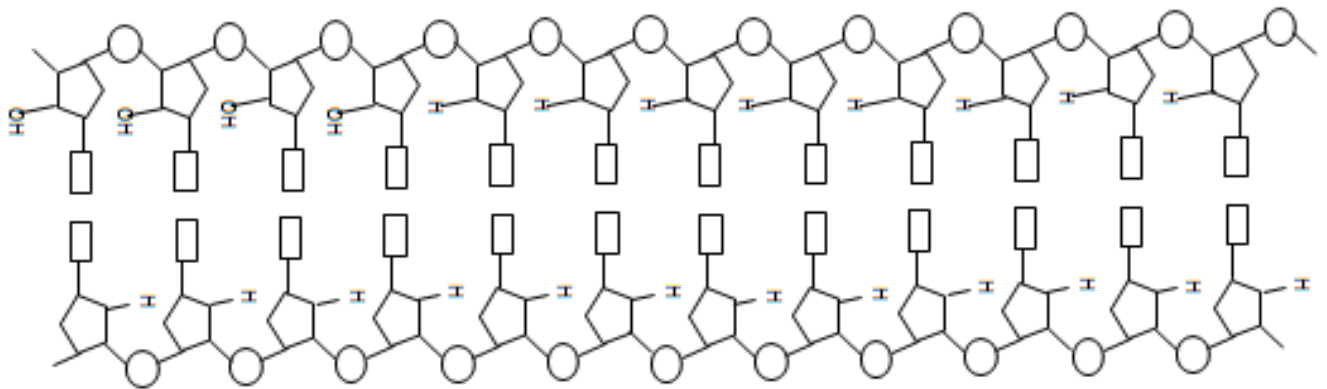


Fig 3.2

(d)(i) Account for the difference in the nucleotides in the top strand compared to bottom strand.

.....

[2]

(ii) Outline the next immediate step in the further processing of nucleic acid strands shown in Fig 3.2.

.....
[1]

[8 marks]

QUESTION 4

A student researcher attempted to clone the Human Growth Hormone (HGH) gene. Two separate samples of HGH DNA were prepared and subjected to agarose gel electrophoresis. Results were shown in Fig.4.1.

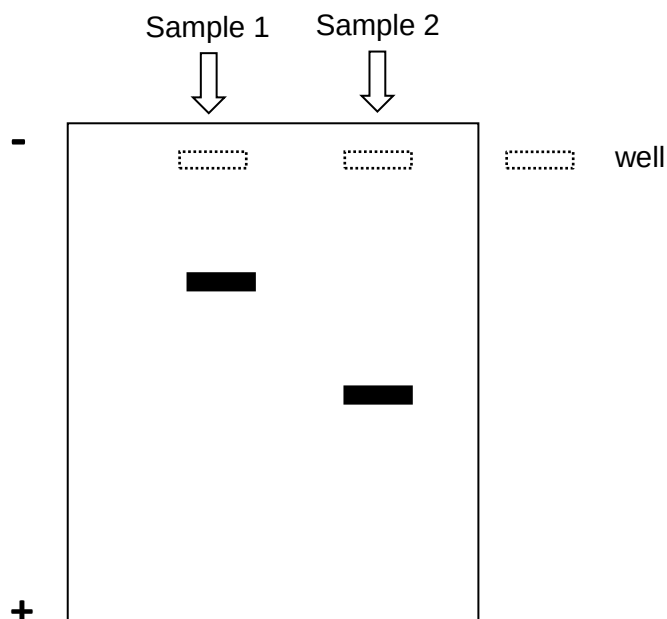


Fig. 4.1

Subsequent cloning procedures involving *Escherichia coli* host cells yield functional HGH protein when Sample 2 was used, but not for Sample 1.

(a) Based on the difference in the bands observed in Fig. 4.1, account for the failure to produce functional HGH protein for Sample 1.

.....

.....

.....

.....[2]

The student researcher has designed the following plasmid (Fig. 4.2) for use in her preliminary trials of a cloning experiment.

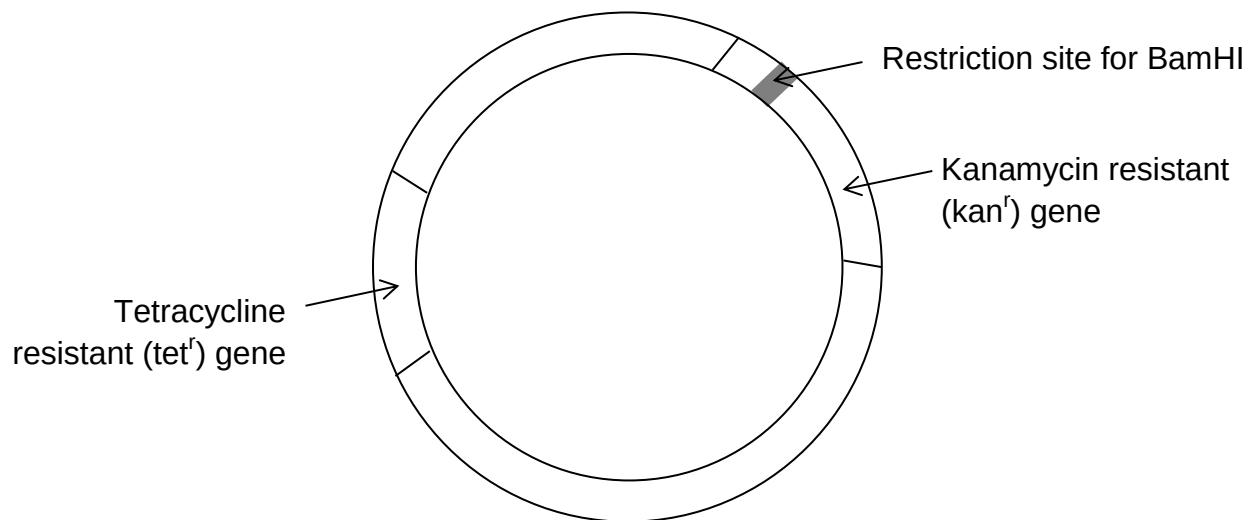


Fig. 4.2

(b) With reference to the properties of bacterial plasmids, provide one reason why this plasmid in Fig 4.2 will not work well as a cloning vector.

.....
[1]

(c) The student researcher has since made improvements to the plasmid in Fig. 4.2 such that it is now a functional cloning vector.

Steps were then taken to create a recombinant plasmid using BamHI and the improved plasmid.

Restriction enzyme	Specific recognition site
BamHI	5'-G [^] G A T C C-3' 3'-C C T A G [^] G-5'

[^] indicates where the restriction enzyme cuts

(i) Describe the natural function of BamHI restriction enzyme.

.....
[1]

(ii) Explain the main problem with using Sample 2 and BamHI in the formation of a recombinant plasmid and suggest how this problem can be solved.

.....

[1½]

(iii) Describe the subsequent steps in the formation of recombinant plasmid.

.....

[2]

(iv) With reference to Fig. 4.2, explain how you would identify transformed bacterial cells that have taken up the recombinant plasmid with the HGH gene using antibiotic selection.

.....

[3]

Escherichia coli is naturally susceptible to ampicillin, which specifically targets cell wall synthesis. The student researcher accidentally spilled ampicillin onto an agar plate containing wild type *E.coli*. Some colonies remained one day after the incident.

(v) Explain the change in allele frequencies in the bacteria population caused by this event.

.....

[2½]

[13 marks]

Section B

Answer **one** question.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

- 5** **(a)** Explain how meiosis and fertilization in sexual reproduction lead to variation within a population. [6]
- (b)** Discuss the importance of genetic variation in natural selection and evolution. [4]
- (c)** Discuss the difficult ethical concerns of the human genome project. [10]

[Total: 20]

- 6** **(a)** Describe the molecular structure of starch (amylose) and cellulose and relate these structures to their functions in living organisms. [7]
- (b)** Contrast Krebs cycle and Calvin cycle. [6]
- (c)** Describe the characteristics of the enzyme active site and the mode of action of enzymes. [7]

[Total: 20]